Dot Products, Rotations, Scaling & Reflections

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Discrete Mathematics

Why Matrices in Computer Graphics?

Every pixel is a coordinate

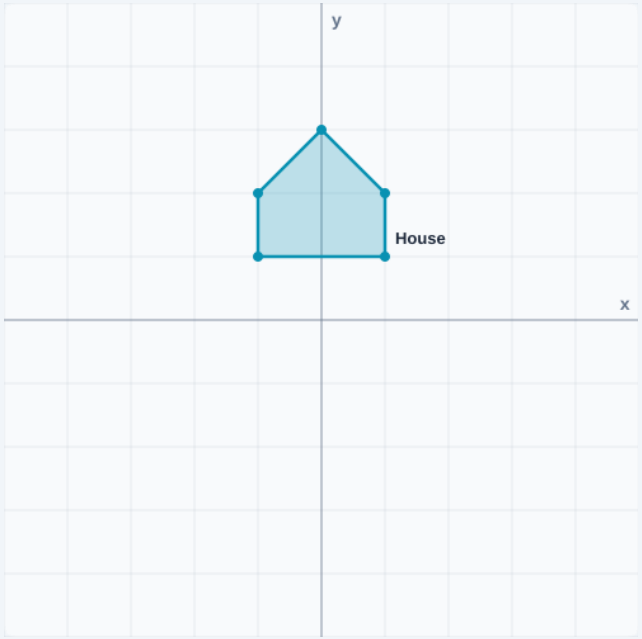
Graphics engines store shapes as lists of (x, y) points called vertices.

Transformations move points

To rotate, scale, or flip a shape, we multiply each vertex by a matrix.

One formula does it all

The dot product inside matrix multiplication is what computes each new coordinate.



A shape = a list of vertices

The Key Idea: Matrix × Vector

```
new point  =  Matrix  ×  old point

[ x' ]     [ a  b ] [ x ]     [ ax + by ]
[ y' ]  =  [ c  d ] [ y ]  =  [ cx + dy ]
```

Each entry in the result is a dot product of a matrix row with the point vector.

$x' = a \cdot x + b \cdot y$
Row 1 dotted with [x, y]

$y' = c \cdot x + d \cdot y$
Row 2 dotted with [x, y]

Rotation Matrix

Rotate by angle θ:

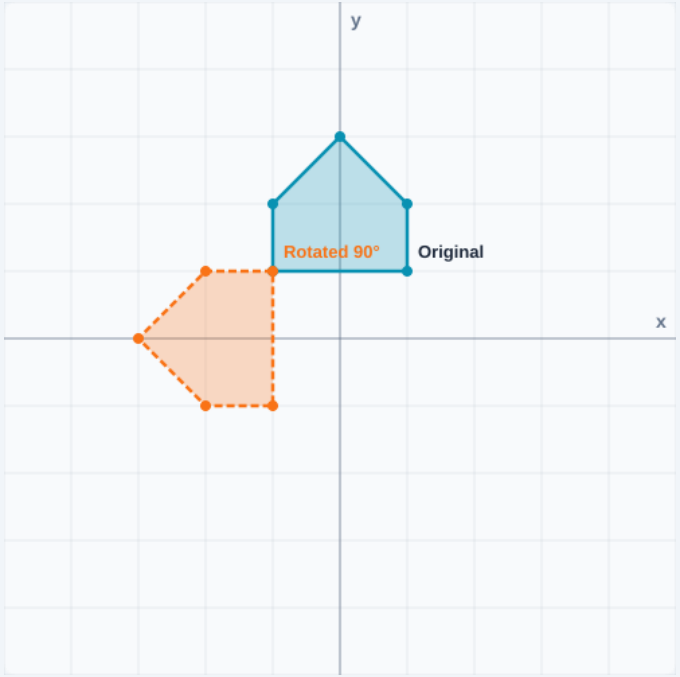
$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Example: Rotate 90° counterclockwise

$\cos 90° = 0, \quad \sin 90° = 1$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

(1,1) becomes (-1,1) ✓



House shape rotated 90° CCW

Scaling Matrix

Scale by factors sx, sy:

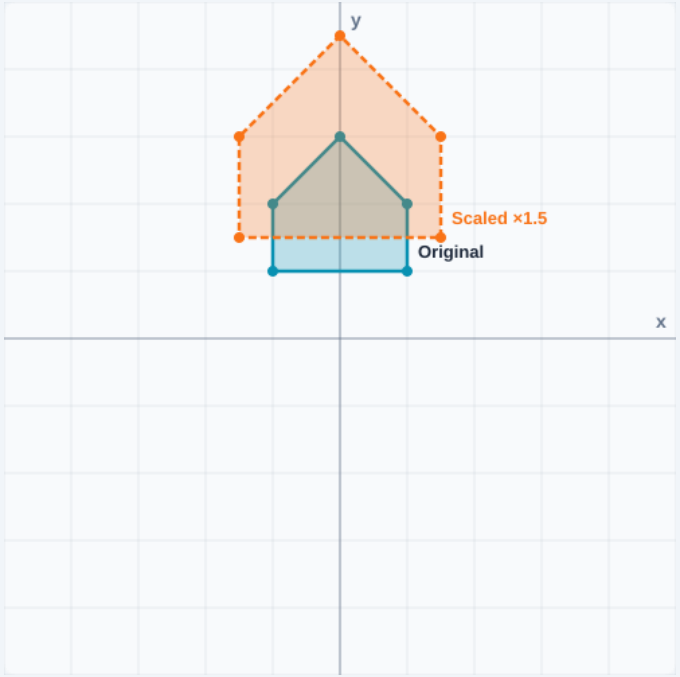
```
[ sx   0 ]
[  0  sy ]
```

Example: Scale by 1.5 in both directions

```
sx = 1.5,  sy = 1.5

[ 1.5   0 ] [ 2 ]   [ 3 ]
[  0  1.5 ] [ 2 ] = [ 3 ]
```

(2,2) becomes (3,3) ✓



House shape scaled by ×1.5

Reflection Matrix

Over y-axis:

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

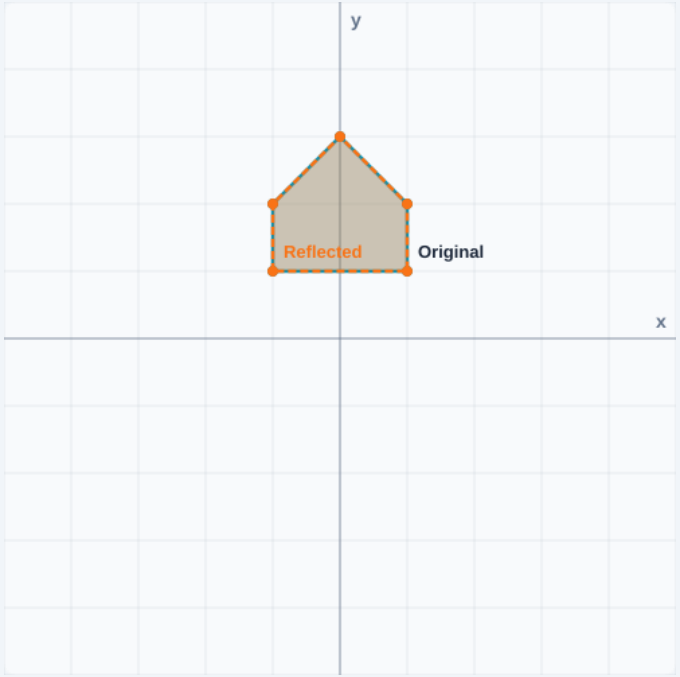
Over x-axis:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Example: Reflect (3, 2) over y-axis

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

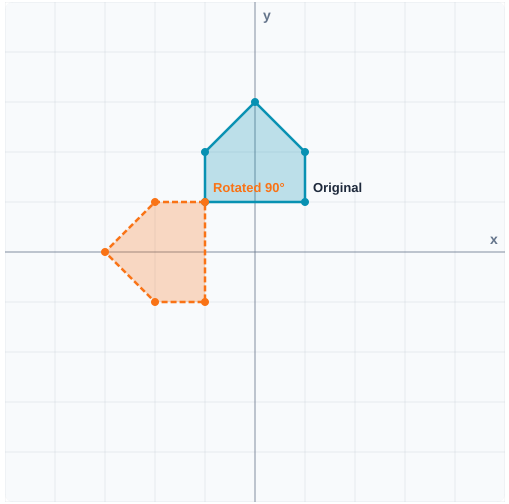
The x flips sign, y stays the same ✓



House reflected over y-axis

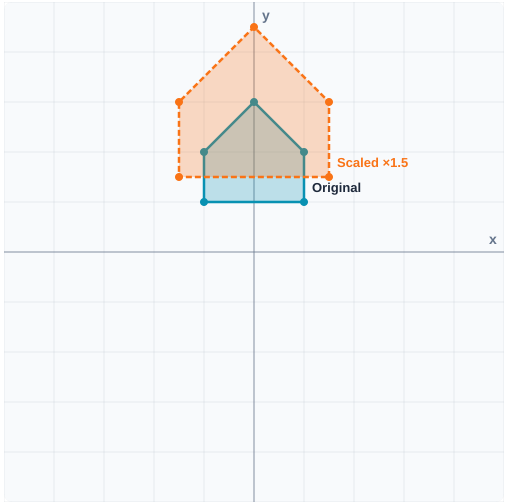
All Three Transformations at a Glance

Rotation



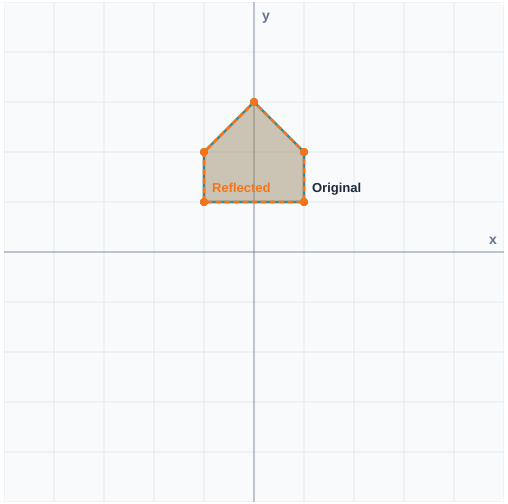
$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Scaling



$$\begin{bmatrix} sx & 0 \\ 0 & sy \end{bmatrix}$$

Reflection



$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

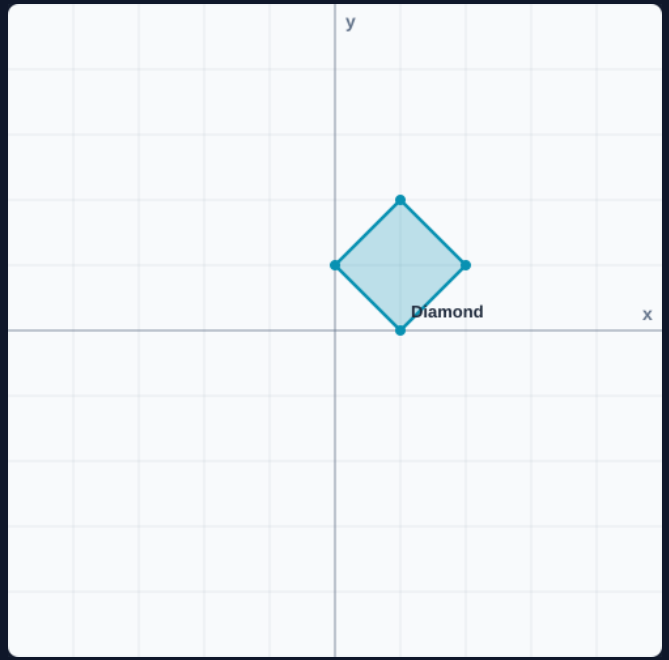
Hands-On Activity: Graph Paper

1. Plot the diamond with vertices:
   (1,0), (2,1), (1,2), (0,1)

2. Apply the 90° rotation matrix to each vertex:
   Multiply [0 -1; 1 0] × each point

3. Apply the scaling matrix [2 0; 0 2] to
   the original vertices

4. Apply the reflection matrix [-1 0; 0 1]
   to the original vertices



Your starting shape

Verify with GeoGebra

Open GeoGebra	Go to geogebra.org/classic or use the classroom app
Enter your matrix	Type: M = {{0,-1},{1,0}} for 90° rotation
Enter a vertex	Type: v = (1, 0) to create a point vector
Compute the product	Type: M * v and observe the transformed point
Compare results	Does your hand calculation match GeoGebra?

Key Takeaways

The dot product is the engine inside every matrix multiplication — each new coordinate is a dot product of a matrix row with the original point.

Rotation, scaling, and reflection are all just 2×2 matrix multiplications — one unified framework for moving shapes.

Composing transformations is just matrix multiplication: rotate then scale = multiply the two matrices.

Computer graphics (games, animation, CAD) use these exact operations millions of times per frame.

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